Math 5010. Foundations of Statistical Theory & Probability

Offers a comprehensive foundational overview of the probability theory and mathematical statistics critical to various methodologies needed to perform and interpret a statistical analysis. Beginning with an exploration of probability distributions, the course explores the theoretical justifications around common statistical measures such as means, variances, and expected values. Likelihood and Bayesian theory are introduced as tools to derive parameter estimates and begin to make inferences. Culminates with formal procedures for statistical inference, including hypothesis testing and the construction of confidence intervals and credible intervals, and calculations of power and sample size. All topics will be grounded in theory, with appropriate practice using a programming language and real-world datasets to communicate and visualize data and results.

Instructor: He Wang

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I am organizing the **Northeastern Applied and Interdisciplinary Mathematics (AIM) Seminar. (https://aimath.sites.northeastern.edu/)**

I also organize the **Data Club (https://aimc.sites.northeastern.edu/)** for MS students in math department.(**information (https://northeastern.instructure.com/courses/229637/pages/data-club-information)**)

Class time and room: 5:50-7:30 pm Ryder Hall 157

The class is in-person and on-ground.

Student Feedback:

- Feel free to **email** me anytime in the semester if you would like to give feedback to me for course improvement. I appreciate your feedback.
- In the end of the semester, you also need to complete the **TRACE** survey for the course for university administration purposes. **https://www.applyweb.com/eval/shibboleth/neu/36892 (https://nam12.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.applyweb.com%2Feval%2Fshibboleth%2Fneu%2F36892&data=04%7C01%7Che.wang%40northeastern.edu%7C4a576e07aa2945a48e2a08d9af7dbc93%7Ca8eec281aaa34daeac9b9a398b9215e7%7C0%7C0%7C637733776853237754%7CUnknown%7CTWFpbGZsb3d8eyJWIjoiMC4wLjAwMDAiLCJQIjoiV2luMzIiLCJBTiI6Ik1haWwiLCJXVCI6Mn0%3D%7C3000&sdata=gN83uXu9dn7oiCHo7s7l17z%2Fb7QNMgwjH0euMfAZBTw%3D&reserved=0)**

Lecture Notes and supporting materials are under Modules

Module 1-Probability Theory Foundations (https://northeastern.instructure.com/courses/229637/modules/1621448)

Module 2- Expectation and Tools (https://northeastern.instructure.com/courses/229637/modules/1621449)

Module 3- Sampling & Limit Theorems (https://northeastern.instructure.com/courses/229637/modules/1621450)

Module 4-Point Estimation & Hypothesis Testing (https://northeastern.instructure.com/courses/229637/modules/1621451)

Module 5- Interval Estimation & Advanced Inference (https://northeastern.instructure.com/courses/229637/modules/1622183)

Class notes, Solutions to Homework, labs, and practices are:

Class notes (https://northeastern.instructure.com/courses/229637/pages/class-notes)

Assignment Solutions (https://northeastern.instructure.com/courses/229637/pages/assignment-solutions)

Practice Questions (https://northeastern.instructure.com/courses/229637/pages/practice-questions)